W2-PM-FINAL | CYBERSECURITY | NETWORKWALKS

PENETRATION TESTING REPORT

FOOTPRINTING & NETWORK SCANNING PHASES

Engagement Field	Details
Pentester Name (Cybersecurity Professional)	**ALEXANDRU DRUMEA**
Program / Batch	**B084J**-Networkwalks
Date	13 November 2026
Modules Completed	W2-PM1 (Multiple Kali Tools) W2-PM2 (Footprinting with Google Hacking Database - GHDB) W2-PM3 (OSINT & Visual Graph Mapping with Maltego) W2-PM4 (Email & Subdomain Harvesting with theHarvester) W2-PM5 (Zenmap Network Scanning)
Client / Target	1. Networkwalks (`networkwalks.com` — secured written permission already) 2. Publicly indexed web assets via GHDB & Microsoft domain (`microsoft.com` passive OSINT) 3. My own local LAN / VirtualBox Host-Only Network (`192.168.56.1/24`)
Permission Secured from Client?	Yes
Phases Covered	Phase 1: Reconnaissance & Footprinting (W2-PM1, W2-PM2, W2-PM3, W2-PM4) Phase 2: Scanning & Network Discovery (W2-PM5) Phase 3–5: In Progress

1. Liability Disclaimer

> I have performed these activities only on the systems & devices where I had secured written permission or the devices/systems that I own myself. All these materials are for education and research purpose only. Do not use anything from here to break the law. The instructor, the authors and Networkwalks are not responsible for what you do with this knowledge. Every action you take is your own responsibility. Misuse can lead to criminal charges, heavy fines, loss of your job and a permanent record. In most countries unauthorised access is a crime even when nothing is damaged.

2. Introduction

This report covers the complete Week 2 practical deliverables of my Cybersecurity & Ethical Hacking internship at Networkwalks (Batch **B084J**). It documents passive and active reconnaissance across five project modules (**W2-PM1** through **W2-PM5**), illustrating how an attacker progresses from collecting publicly available intelligence to discovering active hosts on a local network.

All commands were executed inside a Kali Linux virtual environment (footprinting and OSINT) and on a Windows 11 host machine using Zenmap and Command Prompt (network scanning). Every section below documents the tools used, exact findings observed, risk evaluation, remediation guidance, and visual evidence.

3. Tools Used

The table below lists each tool and platform used across the five project modules in this report and its purpose.

Tool	Purpose
Kali Linux & Windows	Operating systems used for reconnaissance, OSINT, and local network scanning
WHOIS	Find domain registration details (registrar, creation/expiry dates, name servers)
nslookup	Resolve the domain name to its IPv4 address using DNS
curl -I	Inspect HTTP response headers, server software, cookies, and API endpoints
wafw00f	Detect whether a Web Application Firewall (WAF) protects the target website
dnsrecon	Enumerate DNS records (SOA, NS, MX, A, TXT, SPF, SRV, and BIND versions)
Google Search & GHDB	Perform advanced Google Dorking to identify exposed IoT cameras and open directories
Maltego Graph (4.14.0)	Perform visual link analysis and OSINT entity mapping on target domains
theHarvester (4.10.1)	Gather open-source intelligence (emails, subdomains, hosts) from public sources
Zenmap (Nmap 7.93 GUI)	Scan the local subnet (`192.168.56.1/24`) to discover live hosts and generate topology
Windows CMD (`ipconfig`)	Identify local network interfaces, IPv4 address, and subnet mask

4. Activities Performed

4.1 Footprinting & Reconnaissance with Kali Tools (W2-PM1)

I conducted reconnaissance against the authorized target domain `networkwalks.com` using five Kali Linux command-line tools to profile domain ownership, DNS infrastructure, web technologies, and perimeter defenses:

- **WHOIS (`whois networkwalks.com`):** Retrieved domain registration records showing registrar **GoDaddy.com, LLC** (WHOIS server `whois.godaddy.com`), creation date `2019-11-06`, expiry date `2027-11-06`, domain transfer/delete locks (`clientDeleteProhibited`, `clientTransferProhibited`), unsigned DNSSEC, and HostGator name servers (`NS6135.HOSTGATOR.COM`, `NS6136.HOSTGATOR.COM`).
- **Nslookup (`nslookup networkwalks.com`):** Resolved `networkwalks.com` via the local DNS forwarder (`10.0.0.1#53`) to its public IPv4 address **`192.232.216.135`**.
- **Curl (`curl -I https://networkwalks.com`):** Inspected HTTP/1.1 `200 OK` headers, identifying the **Apache** web server, WordPress caching (`X-nginx-cache: WordPress`), session cookie `__wpdm_client` (`Secure` and `HttpOnly`), and exposed WordPress REST API endpoints (`/wp-json/` and `/wp-json/wp/v2/pages/53`).
- **Wafw00f (`wafw00f networkwalks.com`):** Probed the target with 2 HTTP requests and confirmed that `https://networkwalks.com` is protected behind **ModSecurity (SpiderLabs) WAF**.
- **DNSRecon (`dnsrecon -d networkwalks.com`):** Enumerated SOA/NS records (`ns6135.hostgator.com` at `50.87.144.87` and `ns6136.hostgator.com` at `192.232.216.131`, both exposing **BIND version** **`9.16.23-RH`**), A and MX records pointing to `192.232.216.135`, Google site verification and SPF TXT records, and 8 SRV autodiscover records pointing to `cpanelemaildiscovery.cpanel.net` on port `443`.

4.2 Footprinting with Google Hacking Database - GHDB (W2-PM2)

Using Exploit-DB's **Google Hacking Database (GHDB)** and advanced Google Search operators (`intitle:`, `inurl:`, `intext:`), I identified publicly indexed devices and open directories:

- **Exposed Live Security Cameras (10 Findings):** Using dorks such as `inurl:control/camerainfo`, `intitle:"webcamXP" inurl:8080`, `intitle:"webcam 7"`, and `inurl:/multi.html intitle:webcam`, I located 10 publicly accessible live camera feeds (including MOBOTIX interfaces such as `http://86.127.212.113/control/userimage.html` and `http://109.206.96.249:8080/`) accessible without authentication.
- **Open Directory Listings / Mathematics PDFs (10 Findings):** Using `intitle:index.of "parent directory" mathematics pdf`, I identified 10 web servers with directory browsing enabled (`Index of /...`), including `https://mathflix.luc.edu/pdfs/Algebra/` and `https://www.jsoftware.com/books/pdf/`.

4.3 OSINT & Visual Entity Mapping with Maltego (W2-PM3)

Using **Maltego Graph (Desktop) 4.14.0**, I built an OSINT investigation graph starting from the domain entity `networkwalks.com`. Running the **To Emails @domain [Search Engine]** transform harvested an email address entity linked to the domain (`info@networkwalks.com`), demonstrating visual entity correlation.

4.4 OSINT Enumeration with theHarvester (W2-PM4)

I launched **theHarvester 4.10.1** in Kali Linux to evaluate passive OSINT collection against `microsoft.com` using public search engines (`-b baidu`). During CLI testing, I inspected available options (`theHarvester -h`) and verified parameter syntax (confirming lowercase `-l 1000` for result limits), documenting supported passive data sources (`crtsh` , `dnsdumpster` , `hackertarget` , `virustotal` , `urlscan`).

4.5 Network Scanning with Zenmap (W2-PM5)

For the local network discovery exercise on Windows, I performed interface identification, host discovery, and topology export:

- **Local Interface Identification (`ipconfig`):** Identified active adapter `Ethernet 2` configured with IPv4 address **`192.168.56.1`** and Subnet Mask **`255.255.255.0`** (`/24`).
- **Host Discovery (`nmap -sn 192.168.56.1/24`):** Executed a **Ping scan** in **Zenmap (Nmap 7.93)** across all 256 IP addresses in `11.72` seconds, discovering **1 active live host**: **`192.168.56.1`** .
- **Topology Mapping & Export:** Opened the **Topology** tab in Zenmap, enabled the **Topology Legend** (verifying the green host indicator for fewer than 3 open ports connected to `localhost`), and saved the network map in PDF format as `Nmap - Zenmap GUI.pdf` .

5. Risk Analysis / Impact

Based on the observations collected across all five modules, the table below summarizes the identified security exposures and their potential impact:

#	Risk / Finding	Evidence / Observation	Potential Impact	Risk Level
1	Web server & CMS technology disclosure	`curl -I` exposed `Server: Apache` , `X-nginx-cache: WordPress` , and `/wp-json/` on `networkwalks.com`	Enables attackers to fingerprint the web stack and research targeted CMS/plugin vulnerabilities	● **Medium**
2	DNS software version & lack of DNSSEC	`dnsrecon` revealed BIND `9.16.23-RH` on HostGator name servers and no DNSSEC records	Exposes DNS daemon version details and leaves DNS queries without cryptographic validation	● **Medium**
3	Unauthenticated IoT cameras indexed via GHDB	Google Dorks (`inurl:control/camerainfo` , `webcamXP`) exposed 10 live camera feeds without login	Severe privacy breach and unauthorized viewing or control of physical surveillance devices	● **Critical**
4	Directory listing enabled on web servers	GHDB dorks (`intitle:index.of "parent directory"`) exposed 10 open file directories	Unauthorized enumeration and bulk downloading of internal or unlinked documents	● **Medium**

#	Risk / Finding	Evidence / Observation	Potential Impact	Risk Level
5	Public email & domain entity exposure	Maltego transform harvested organizational email entity linked to `networkwalks.com`	Exposed corporate email addresses can be targeted in phishing or social engineering campaigns	● Low
6	WAF & Server IP identification	`nslookup` resolved `192.232.216.135` and `wafw00f` identified `ModSecurity (SpiderLabs)`	Reveals hosting IP and defensive perimeter architecture, aiding targeted evasion analysis	● Low
7	Local subnet host visibility	Zenmap Ping Scan (`nmap -sn 192.168.56.1/24`) discovered active host `192.168.56.1`	Demonstrates how ICMP/ARP discovery enumerates live endpoints on an internal LAN segment	● Low

Risk level key: ● Critical | ● Medium | ● Low

Disclaimer: The risks above are reconnaissance and scanning observations gathered strictly within authorized educational scope, not confirmed exploitable vulnerabilities. No exploitation was performed.

6. Recommendations

- **Minimize HTTP Header & API Exposure:** Configure Apache/Nginx to suppress verbose server banners (`ServerTokens Prod`) and restrict public access to unnecessary WordPress REST API (`/wp-json/`) endpoints.
- **Harden DNS Infrastructure & Enable DNSSEC:** Disable DNS software version disclosure (`version "none";` in BIND) on authoritative name servers and deploy DNSSEC to prevent DNS spoofing.
- **Restrict Access to IoT & IP Cameras:** Require strong authentication, enforce HTTPS, and place IP camera interfaces behind firewalls or VPNs rather than exposing them directly to search engine crawlers.
- **Disable Web Server Directory Browsing:** Configure web servers (`Options -Indexes` in Apache) so `Index of /` directory listings are disabled globally.
- **Maintain & Tune WAF Rules:** Keep **ModSecurity WAF** enabled and updated with the latest OWASP Core Rule Set (CRS) to filter automated reconnaissance probes.
- **Protect Organizational Email Addresses:** Enforce SPF, DKIM, and DMARC policies alongside anti-phishing filtering and staff security awareness training.
- **Perform Regular Internal Network Audits:** Periodically scan local subnets with Nmap/Zenmap to maintain an accurate host inventory and detect unauthorized endpoints.

7. Conclusion

During Week 2 of my Cybersecurity & Ethical Hacking internship at Networkwalks (Batch **B084J**), I completed all five practical modules covering passive footprinting, OSINT, and active network scanning (**W2-PM1** through **W2-PM5**).

Across **W2-PM1**, **W2-PM3**, and **W2-PM4**, I applied command-line and visual OSINT tools (`whois` , `nslookup` , `curl` , `wafw00f` , `dnsrecon` , `Maltego` , and `theHarvester`) to map domain registration records, IP infrastructure, DNS services, WAF protections, and email entities. In **W2-PM2**, I demonstrated how Google Hacking Database dorks uncover misconfigured directory listings and unauthenticated live cameras. Finally, in **W2-PM5**, I used Windows `ipconfig` and `Zenmap` to scan my `192.168.56.1/24` network, verify live hosts, and export a visual network topology.

These exercises reinforced that structured reconnaissance and clear technical documentation are essential to cybersecurity assessments, and must always be conducted strictly within authorized legal boundaries.

8. Evidences Collected

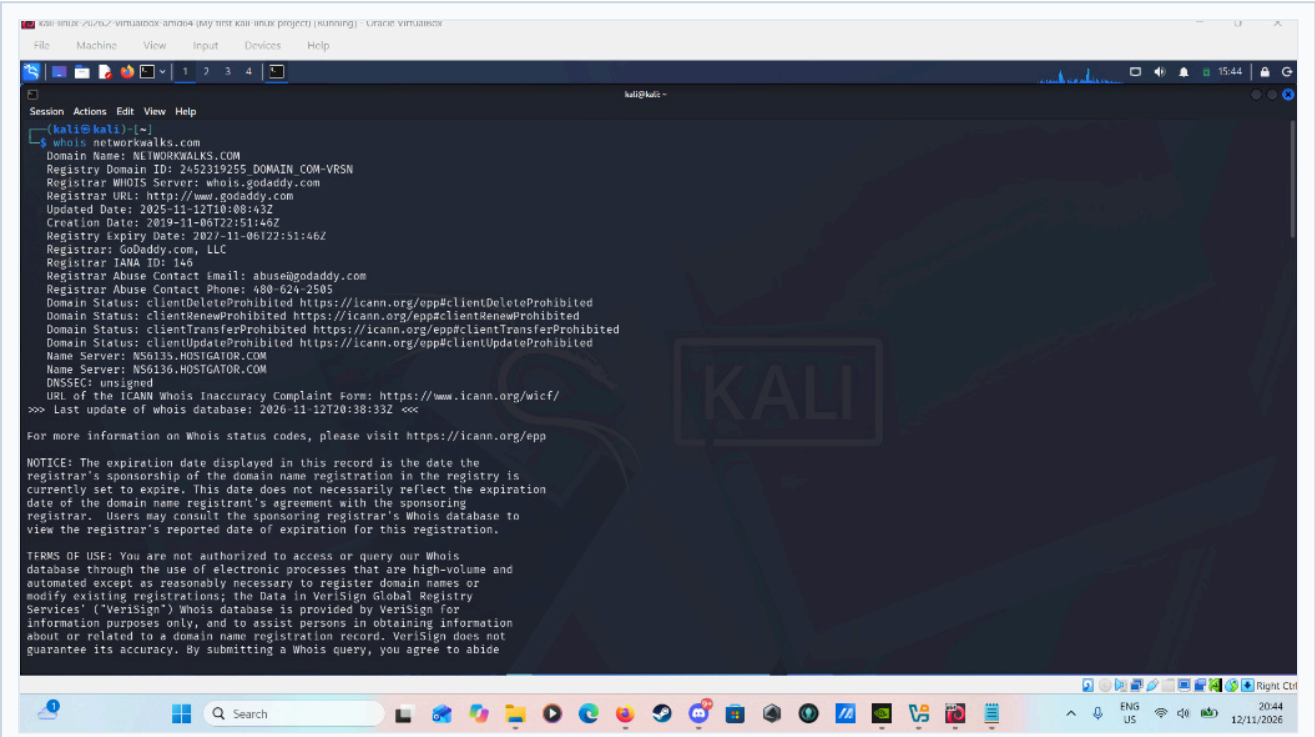


Figure 1 (W2-PM1): WHOIS domain registration and HostGator name server lookup for networkwalks.com.



Figure 2 (W2-PM1): DNS resolution via nslookup identifying IPv4 address 192.232.216.135.

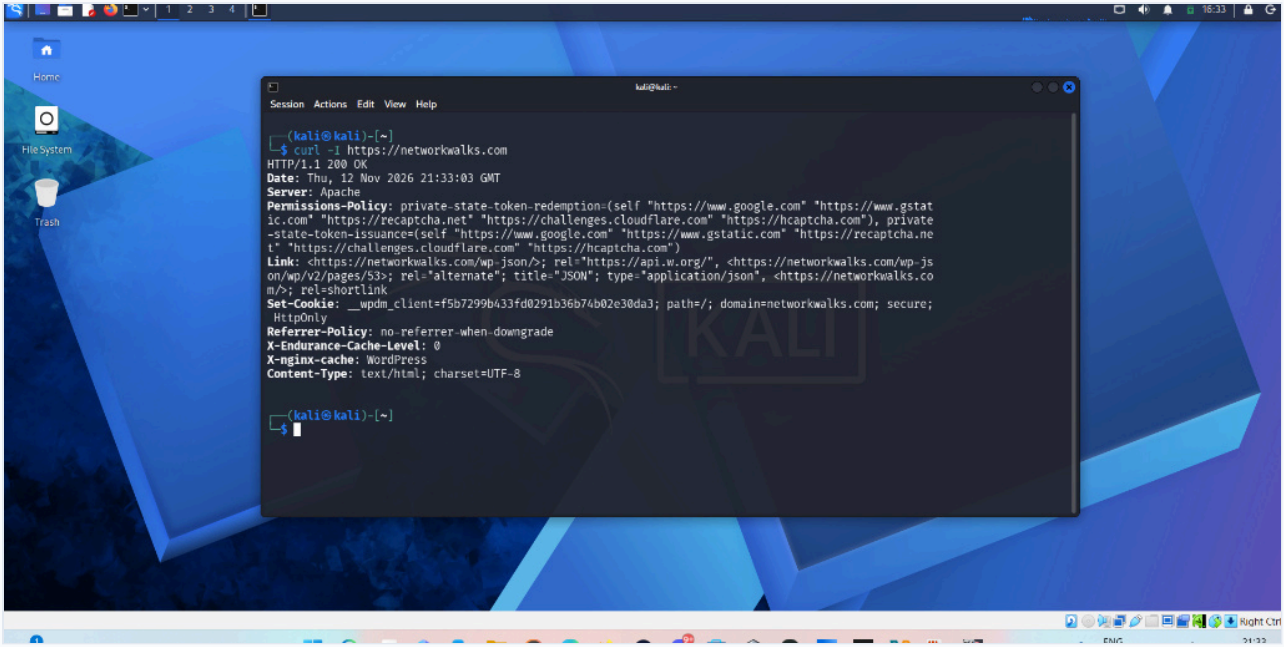


Figure 3 (W2-PM1): HTTP response headers via curl -I revealing Apache server, WordPress cache, and /wp-json/ endpoints.

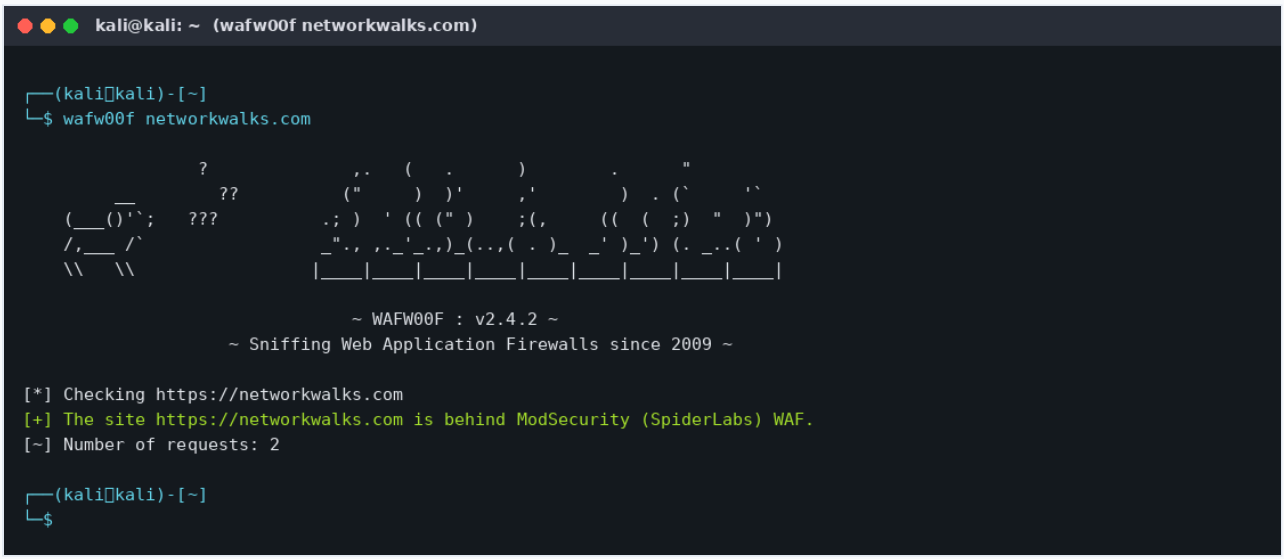


Figure 4 (W2-PM1): Web Application Firewall fingerprinting via wafw00f detecting ModSecurity (SpiderLabs).

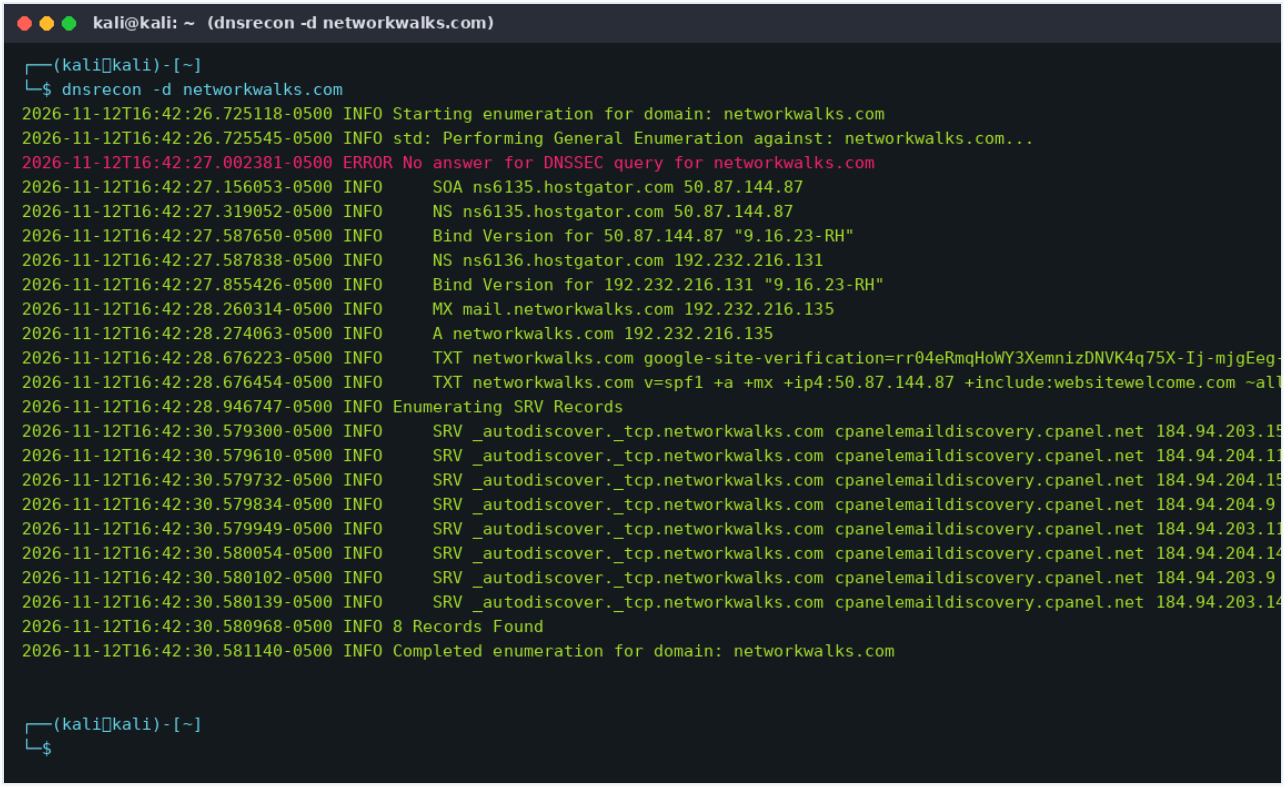


Figure 5 (W2-PM1): DNS enumeration via dnsrecon showing SOA, NS (BIND 9.16.23-RH), MX, TXT/SPF, and 8 SRV records.

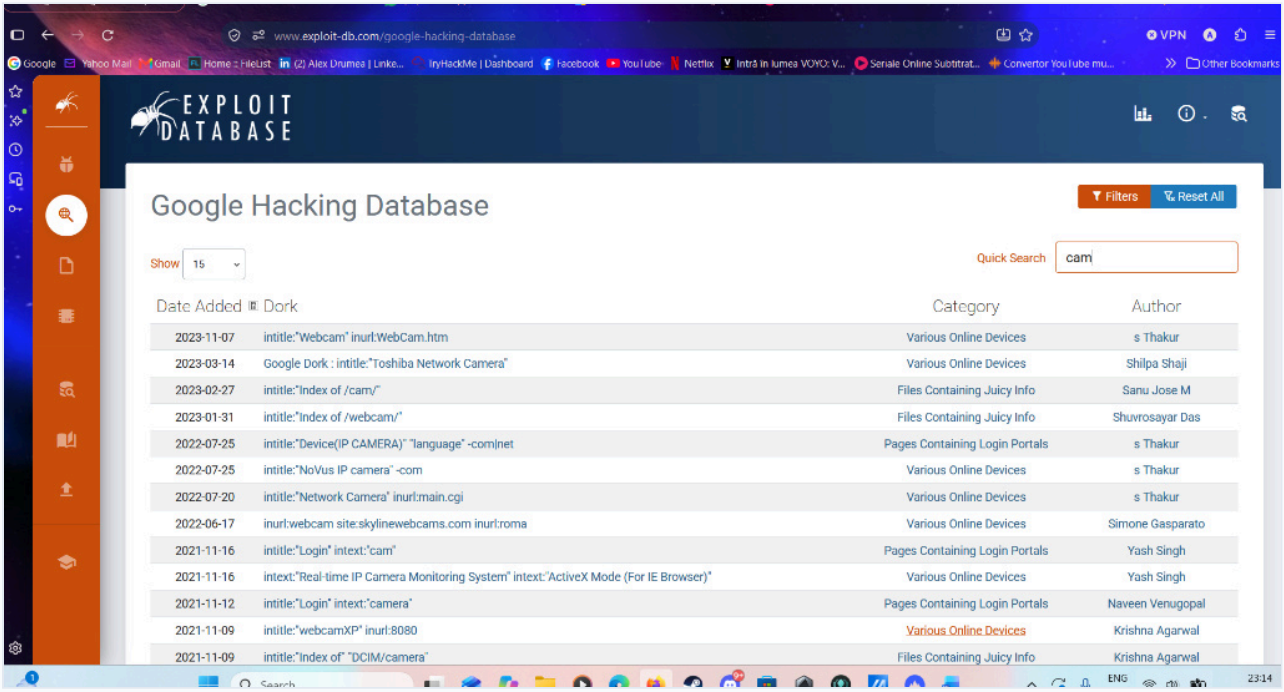


Figure 6 (W2-PM2): Exploit-DB Google Hacking Database (GHDB) query exploration for exposed devices.

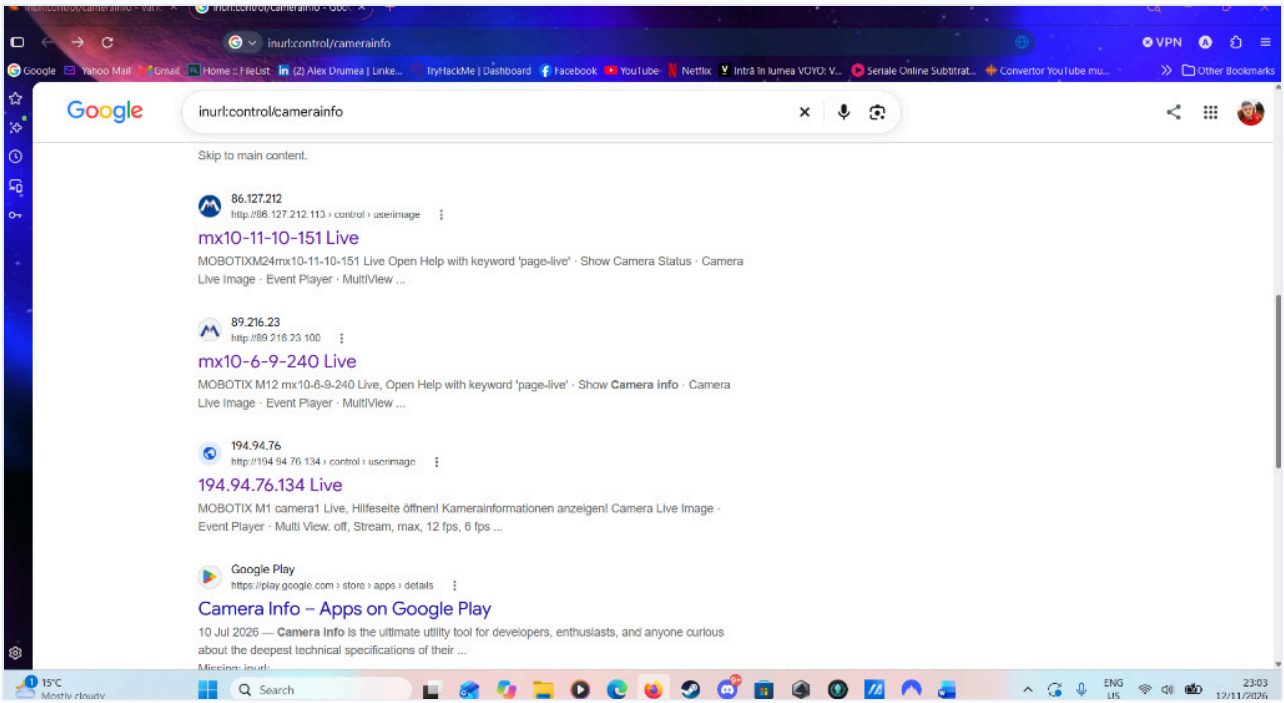


Figure 7 (W2-PM2): Google Dork search (inurl:control/camerainfo) returning publicly indexed MOBOTIX IP cameras.

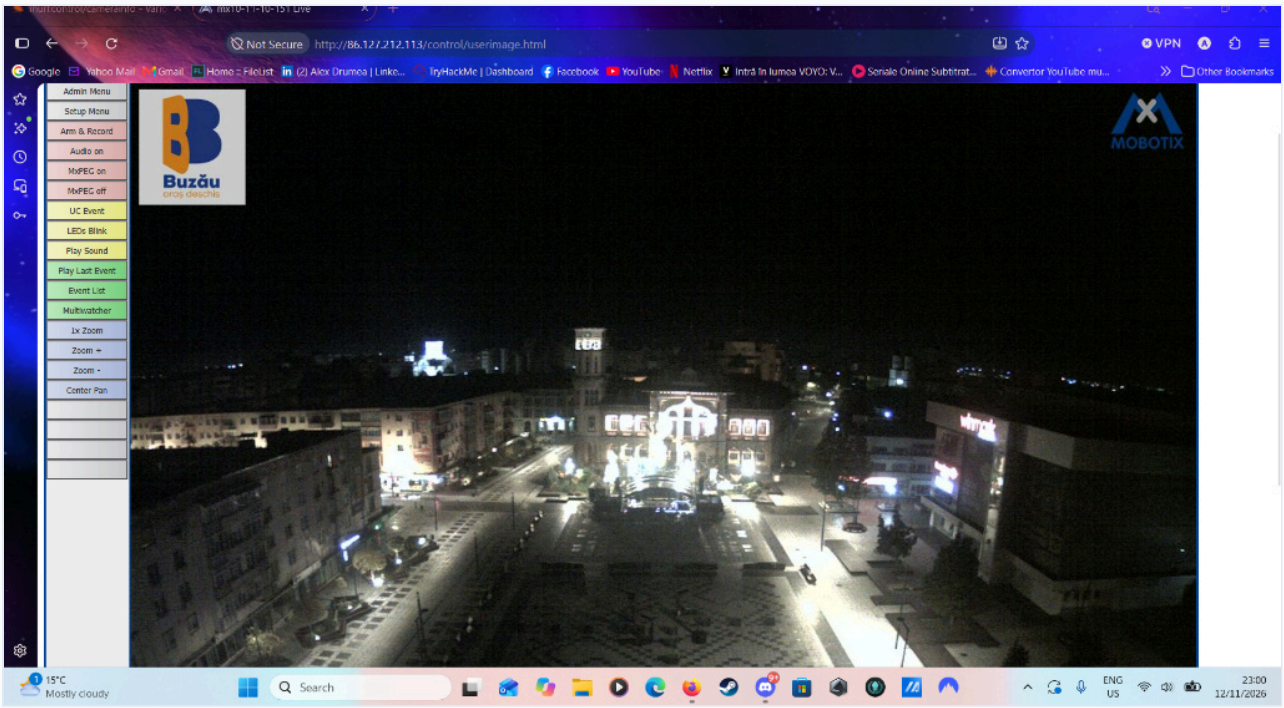


Figure 8 (W2-PM2): Unauthenticated live MOBOTIX camera interface discovered via GHDB dorking.

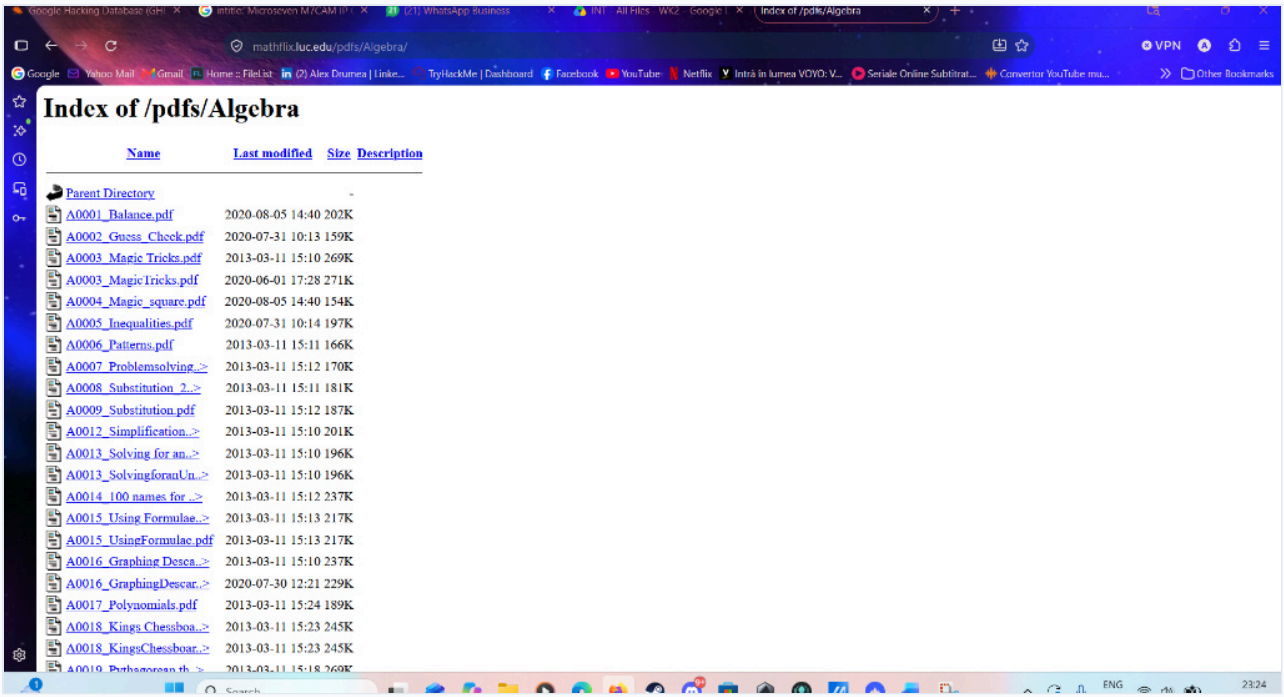


Figure 9 (W2-PM2): Open directory listing (Index of /pdfs/Algebra) identified using GHDB dorks.

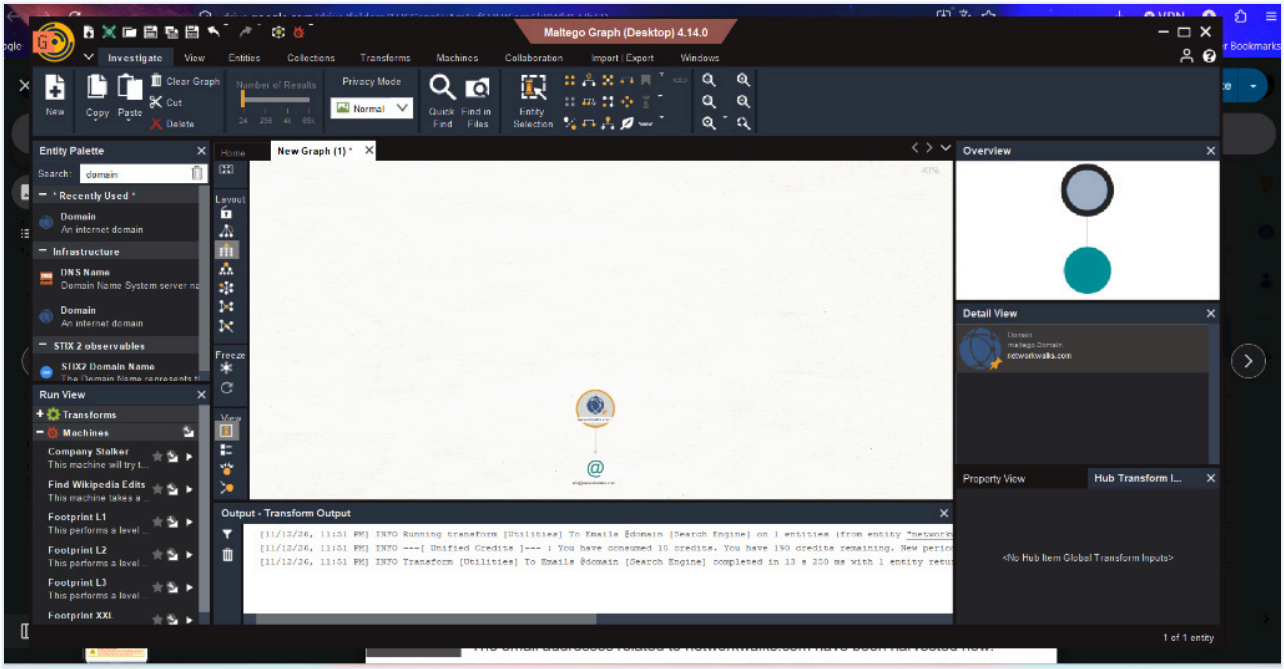


Figure 10 (W2-PM3): Maltego Graph 4.14.0 visual link analysis mapping networkwalks.com to an associated email entity.

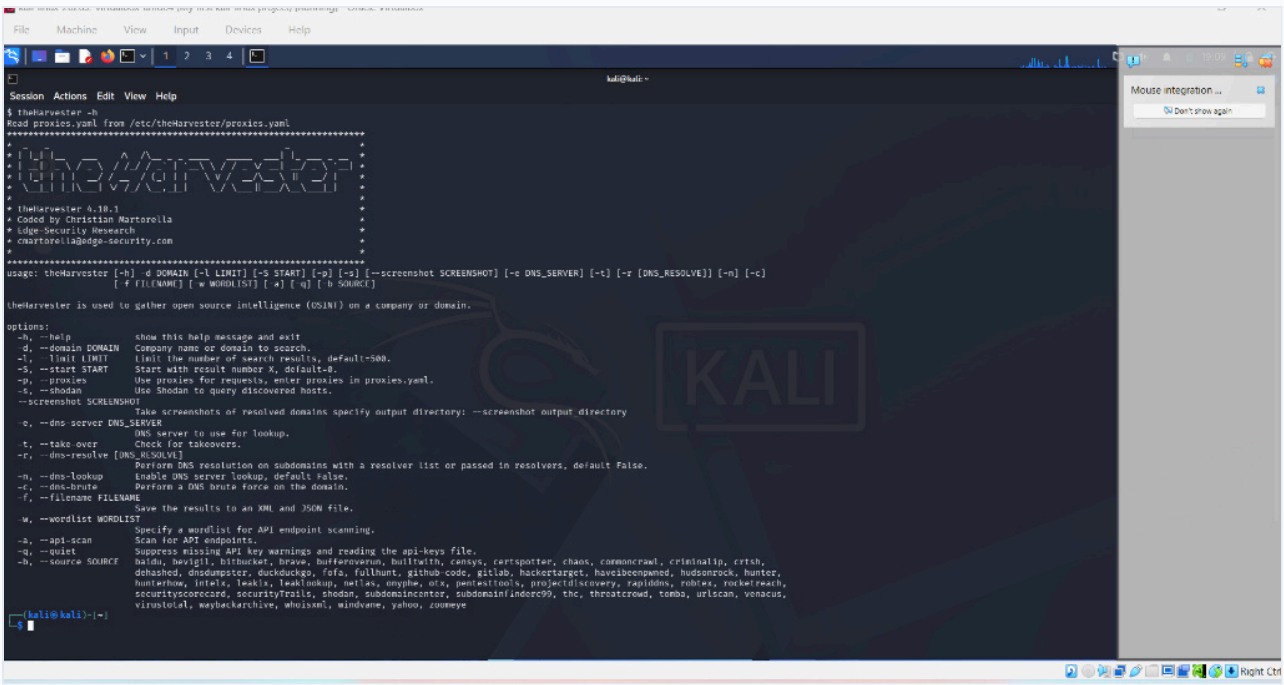


Figure 11 (W2-PM4): theHarvester 4.10.1 CLI options and passive OSINT data sources in Kali Linux.

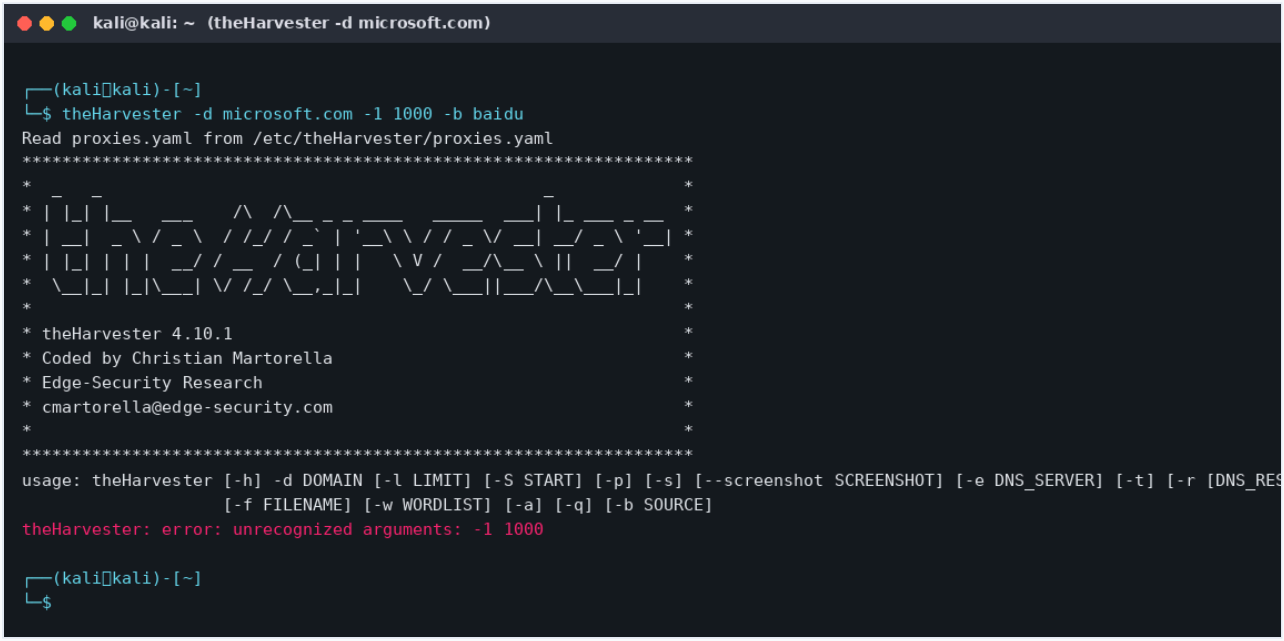


Figure 12 (W2-PM4): theHarvester CLI parameter verification targeting microsoft.com.

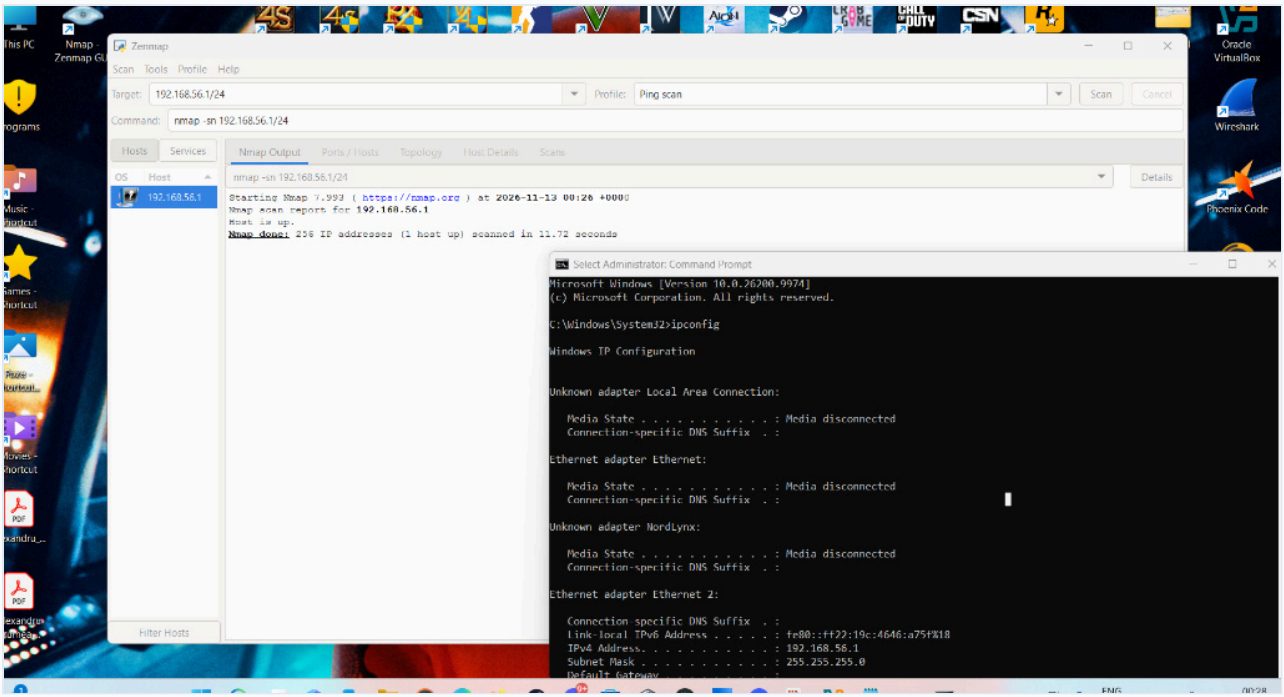


Figure 13 (W2-PM5): Windows ipconfig (192.168.56.1/24) and Zenmap Ping scan (nmap -sn 192.168.56.1/24) identifying 1 active host.

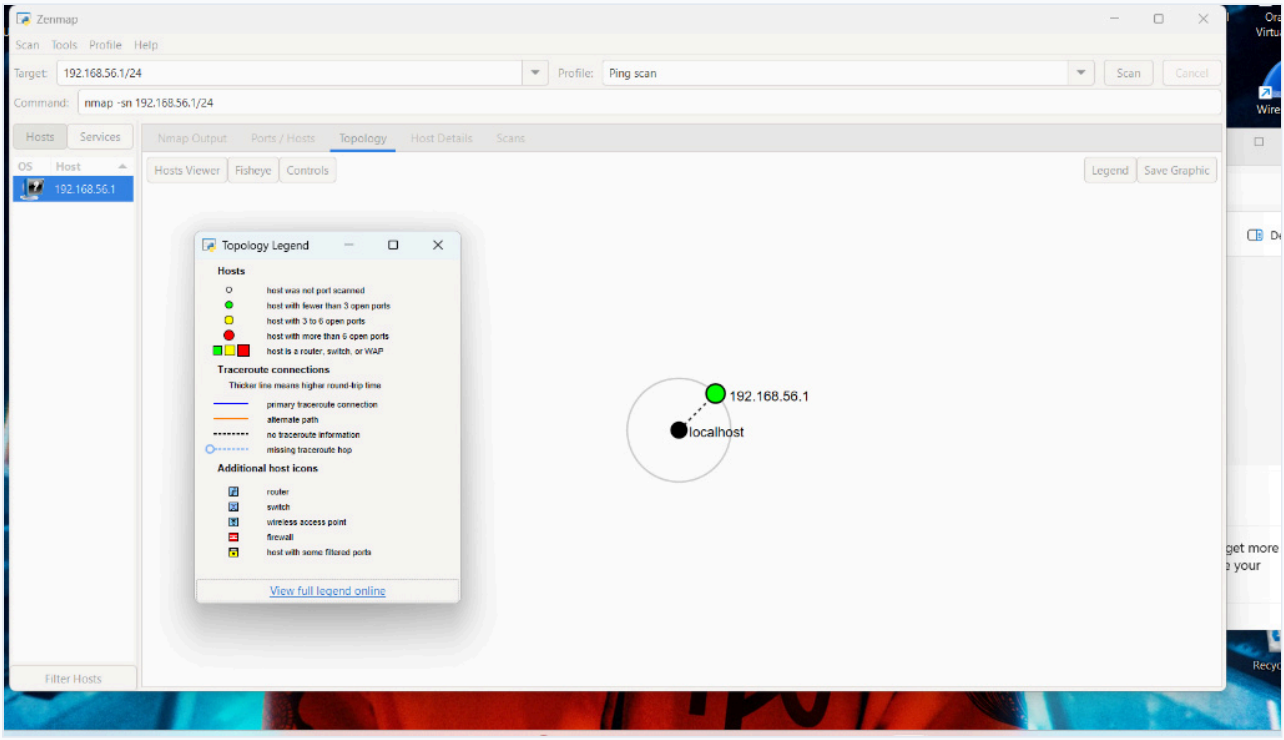


Figure 14 (W2-PM5): Zenmap network topology graph with Topology Legend enabled.

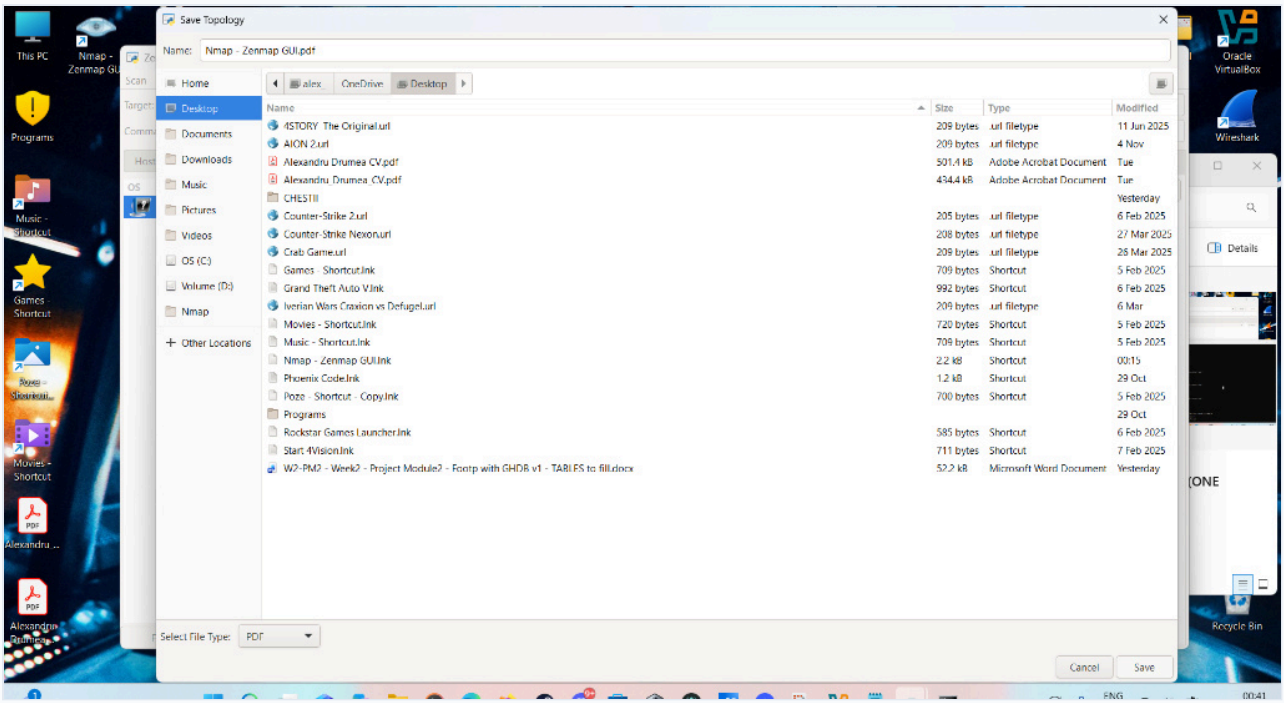


Figure 15 (W2-PM5): Exporting the Zenmap network topology diagram to PDF format (Nmap - Zenmap GUI.pdf).

-End-

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Project Information: Cybersecurity Program at Networkwalks | Week: 02 | Modules: W2-PM1 to W2-PM5